■ 1.8.4 Redrawing and Combining Plots

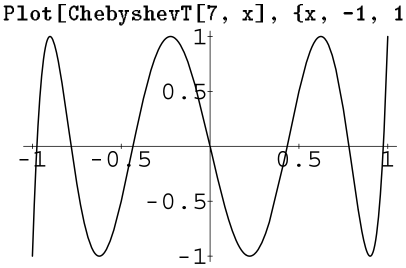
Mathematica saves information about every plot you produce, so that you can later redraw it. When you redraw plots, you can change some of the options you use.

`Show[`*plot*`]`	redraw a plot
`Show[`*plot*`, `*option*`->`*value*`]`	redraw with options changed
`Show[`$plot_1$`, `$plot_2$`, `...`]`	draw several plots together
`Options[`*plot*`]`	show the options used for a particular plot
`InputForm[`*plot*`]`	show the information that is saved about a plot

Functions for manipulating plots.

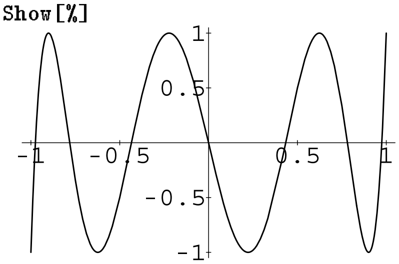
Here is a simple plot. `-Graphics-` is usually printed on the output line to stand for the information that *Mathematica* saves about the plot.

In[1]:= **`Plot[ChebyshevT[7, x], {x, -1, 1}]`**



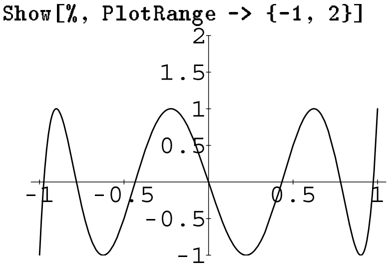
This redraws the plot from the previous line.

In[2]:= **`Show[%]`**



When you redraw the plot, you can change some of the options. This changes the choice of y scale.

In[3]:= **`Show[%, PlotRange -> {-1, 2}]`**



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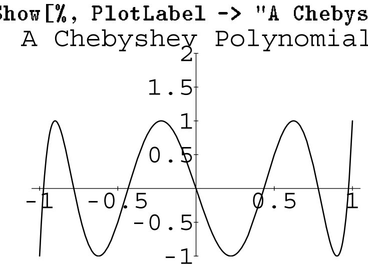
This takes the plot from the previous line, and changes another option in it.

```
In[4]:= Show[%, PlotLabel -> "A Chebyshev Polynomial"]
```

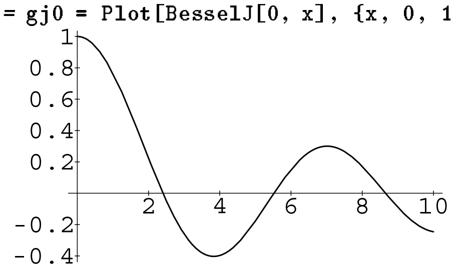


By using `Show` with a sequence of different options, you can look at the same plot in many different ways. You may want to do this, for example, if you are trying to find the best possible setting of options.

You can also use `Show` to combine plots. It does not matter whether the plots have the same scales: *Mathematica* will always choose new scales to include the points you want.

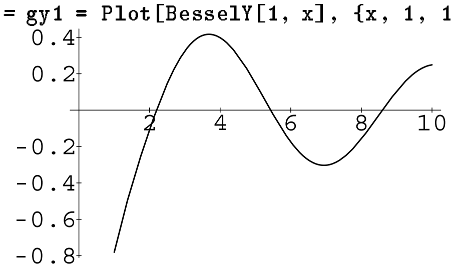
This sets `gj0` to be a plot of $J_0(x)$ from $x = 0$ to 10.

```
In[5]:= gj0 = Plot[BesselJ[0, x], {x, 0, 10}]
```



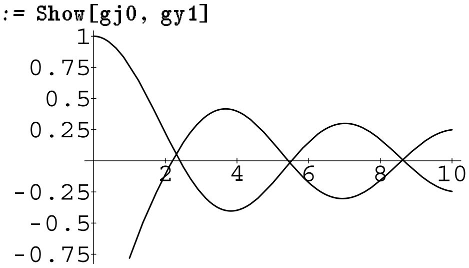
Here is a plot of $Y_1(x)$ from $x = 1$ to 10.

```
In[6]:= gy1 = Plot[BesselY[1, x], {x, 1, 10}]
```



This shows the previous two plots combined into one. Notice that the x scale in the combined plot is chosen to include all the points in both the plots.

```
In[7]:= Show[gj0, gy1]
```



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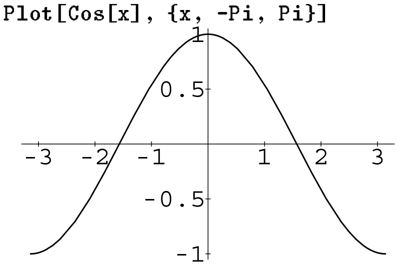
When you make a plot, *Mathematica* saves the list of points it used, together with some other information. Using what is saved, you can redraw plots in many different ways with `Show`. However, you should realize that whatever options you specify, `Show` still has the same basic set of points to work with. So, for example, if you set the options so that *Mathematica* displays a small portion of your original plot magnified, you will probably be able to see the individual sample points that `Plot` used. Options like `PlotPoints` can only be set in the original `Plot` command itself. (*Mathematica* always plots the acutal points it has; it avoids using smoothed or splined curves, which can give misleading results in mathematical graphics.)

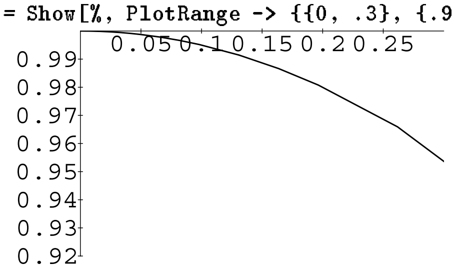
Here is a simple plot.

In[8]:= **`Plot[Cos[x], {x, -Pi, Pi}]`**



This shows a small region of the plot in a magnified form. At this resolution, you can see the individual line segments that were produced by the original `Plot` command.

In[9]:= **`Show[%, PlotRange -> {{0, .3}, {.92, 1}}]`**



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